

Introduction to Microeconomics
Lectures 3 & 4—Supply & Demand

Prof. Alejandro Melo-Ponce



Years
Anniversary

ECON101
Spring 2026

- Why do male soccer players earn more than female soccer players?
- Why do diamonds cost more than water?
- Why do Picasso's paintings sell for more than other artists?
- Is it the cost of production that determines prices, or is it willingness to pay?
- Alfred Marshall (1890) was the first to clearly explain how both costs and willingness to pay interact to determine market prices.

Definition: Market

The market for any good or service consists of all (actual or potential) buyers or sellers of that good or service.

The supply–demand framework

- A *market* is not necessarily a physical place.
 - Mechanism through which buyers and sellers interact.
- The supply–demand model is a *workhorse* of microeconomics.
- Prices and quantities: **endogenous variables**.
- Preferences, technology, income, and other determinants: **exogenous variables** (*ceteris paribus*).
- Focus is on *single markets*, or at most a few interrelated but mostly independent markets.
 - This is the **partial equilibrium** approach.

The demand function

- Demand is represented by a function:

$$q^D = D(p; \alpha)$$

- It gives the *total quantity* of a good or service that buyers wish to buy.
 - As a function of its own price p .
 - And of other factors α (preferences, income, prices of related goods, expectations, etc.).
- The *demand curve* is a **ceteris paribus** analysis.
 - Fix the determinants.
 - Analyze how quantity demanded changes with respect to its own price.

From demand function to demand curve

- Fix a level of determinants: $\bar{\alpha}$.
- Price p : **exogenous variable**.
- Quantity demanded q^D : **endogenous variable**.
- The demand curve is the *inverse demand*:

$$p = D^{-1}(q; \bar{\alpha})$$

- It represents the pairs (q, p) at which buyers are willing to purchase the good.

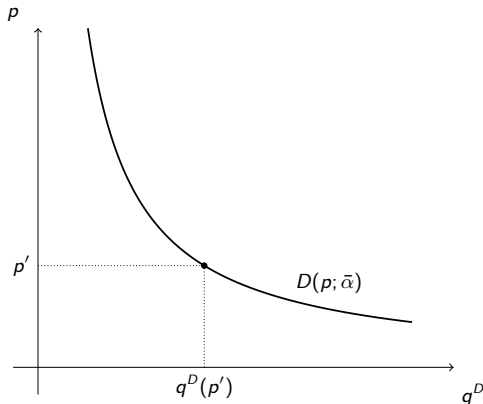
Law of Demand

- As price increases:
 - the good becomes less affordable,
 - buyers substitute toward cheaper alternatives.
- Quantity demanded decreases.

Interpreting the demand curve

Horizontal interpretation: at price p , how much quantity is demanded?

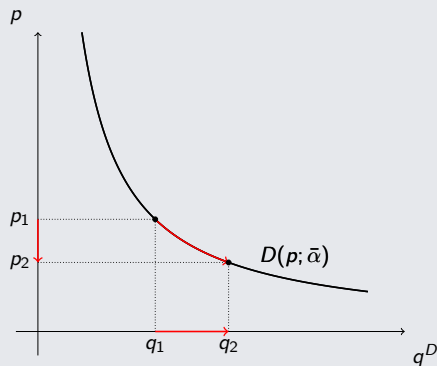
Vertical interpretation: at quantity q , what is the buyer's **willingness to pay**?



Movements along the demand curve and shifts of the demand schedule

Changes in quantity demanded

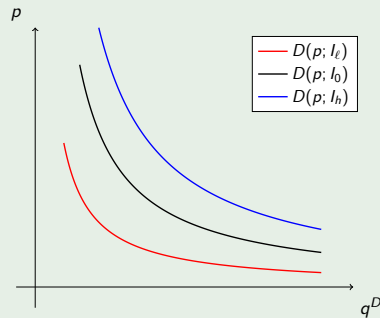
A price change from p_1 to p_2 induces a change in *quantity demanded* from q_1 to q_2 . This is a movement along the demand curve.



Shifts of Demand

Suppose there is a change in $\bar{\alpha} \rightarrow \hat{\alpha}$. This corresponds to a **shift of the demand curve**.

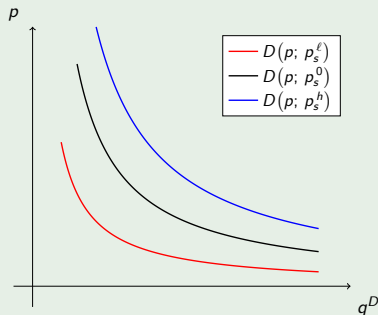
(i) Changes in income: Consider $0 < I_\ell < I_0 < I_h$.



Price of substitutes and complements

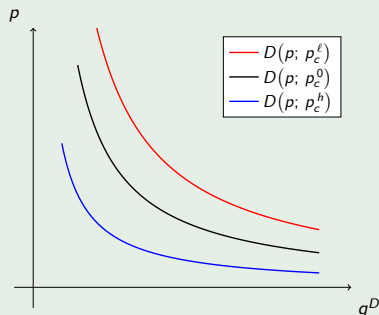
Shifts of Demand: Price of a Substitute

(ii) Consider the following prices of a substitute good: $0 < p_s^\ell < p_s^0 < p_s^h$.



Shifts of Demand: Price of a Complement

(iii) Consider the following prices of a complement good: $0 < p_c^\ell < p_c^0 < p_c^h$.



The supply function

- Supply is represented by a function:

$$q^S = S(p; \gamma)$$

- It gives the *total quantity* of a good or service that sellers wish to sell.
 - As a function of its own price p .
 - And of other factors γ (input costs, technology, number of sellers, expectations, etc.).
- The *supply curve* is a **ceteris paribus** analysis.
 - Fix the determinants.
 - Analyze how quantity supplied changes with respect to its own price.

From supply function to supply curve

- Fix a level of determinants: $\bar{\gamma}$.
- Price p : **exogenous variable**.
- Quantity supplied q^S : **endogenous variable**.
- The supply curve is the *inverse supply*:

$$p = S^{-1}(q; \bar{\gamma})$$

- It represents the pairs (q, p) at which sellers are willing to offer the good for sale.

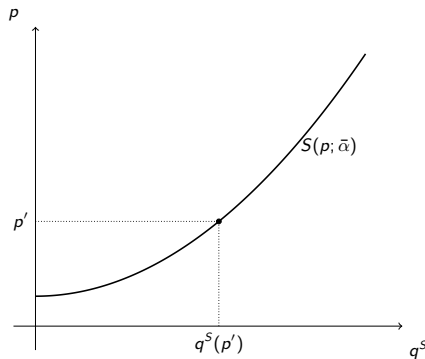
Law of Supply

- As price increases:
 - producing/selling more becomes worthwhile,
 - marginal (opportunity) cost is increasing.
- Quantity supplied increases.

Interpreting the supply curve

Horizontal interpretation: at price p , how much quantity is supplied?

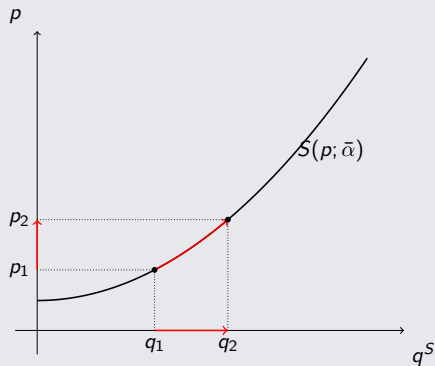
Vertical interpretation: at quantity q , what is the **minimum price** sellers are willing to accept?



Movements along the supply curve and shifts of the supply schedule

Changes in quantity supplied

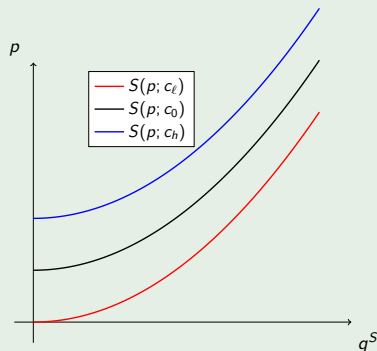
A price change from p_1 to p_2 induces a change in *quantity supplied* from q_1 to q_2 . This is a movement along the supply curve.



Shifts of Supply

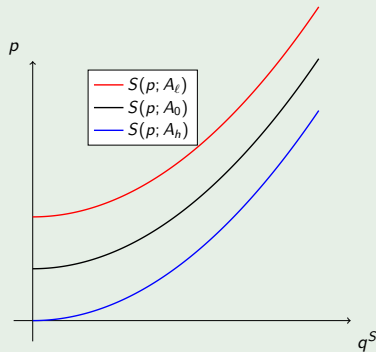
Suppose there is a change in $\bar{\gamma} \rightarrow \hat{\gamma}$. This corresponds to a **shift of the supply curve**.

(i) Changes in input costs: Consider $0 < c_\ell < c_0 < c_h$.



Shifts of Supply

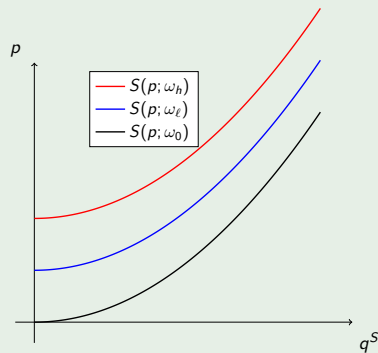
(ii) Technological change: Consider $A_\ell < A_0 < A_h$.



Shifts of Supply

Suppose there is a natural disaster affecting production conditions.

(iii) Natural shock: Consider $\omega_0 < \omega_\ell < \omega_h$.



Consumer Surplus

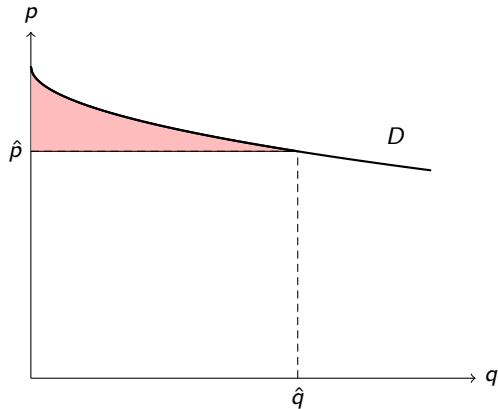
Consumer Surplus is the excess of the benefit received from a good over the amount paid for it. We can calculate consumer surplus as the marginal benefit (or value) of a good minus its price, summed over the quantity bought. It is measured by the area under the demand curve and above the price paid, up to the quantity bought.

Producer Surplus

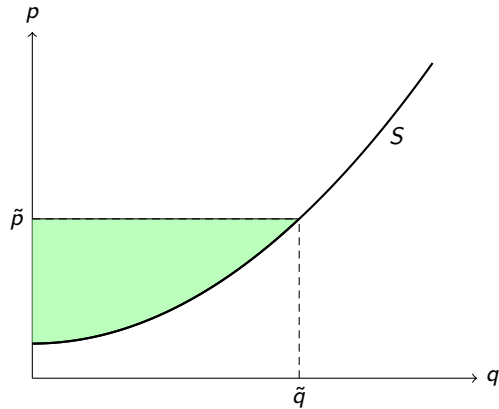
Producer Surplus is the excess of the amount received from the sale of a good over the cost of producing it. We calculate it as the price received for a good minus the minimum-supply price (marginal cost), summed over the quantity sold. On a graph, producer surplus is shown by the area below the price received and above the supply curve, summed over the quantity sold.

Consumer and Producer Surplus

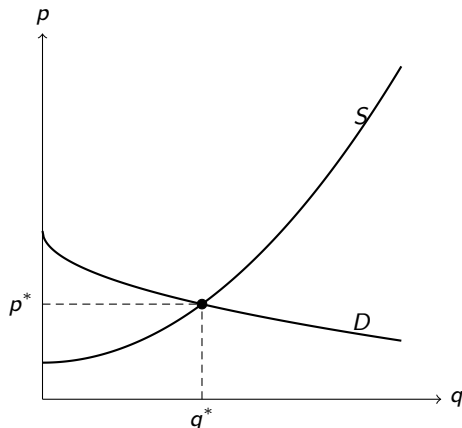
Consumer Surplus



Producer Surplus



Market Equilibrium



Market equilibrium

Equilibrium occurs at the price–quantity pair for which both buyers and sellers are satisfied.

For a fixed level of the determinants of demand and supply, α and γ , a **market equilibrium** is a pair

$$(q^*, p^*)$$

such that

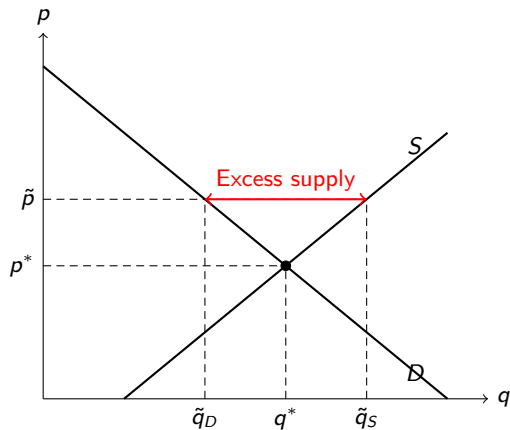
$$q^* = D(p^*; \alpha),$$

$$q^* = S(p^*; \gamma).$$

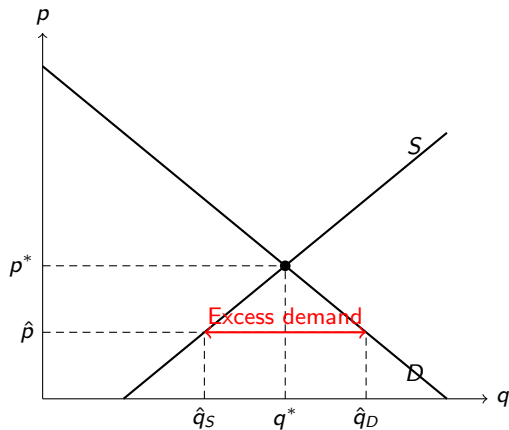
Equilibrium is a *consistency condition*: individual plans are mutually compatible at (q^*, p^*) .

- Buyers are willing to buy q^* units at price p^* .
- Sellers are willing to sell q^* units at price p^* .

Excess Supply and Excess Demand



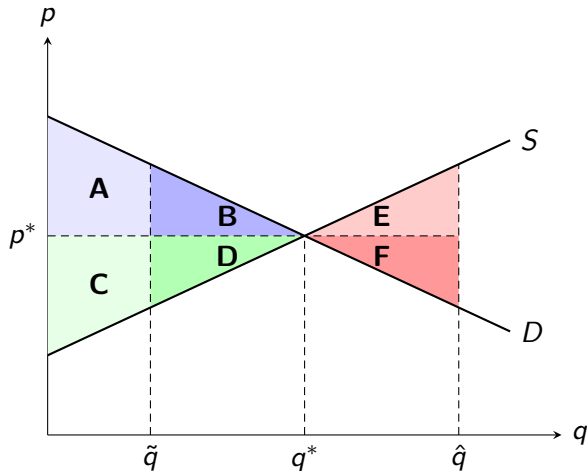
$$\text{Excess Supply}(\tilde{p}) = \tilde{q}_S - \tilde{q}_D$$



$$\text{Excess Demand}(\hat{p}) = \hat{q}_D - \hat{q}_S$$

Social Surplus

Social surplus is the sum of producer and consumer surplus. It represents the total gains from participating in a market.



Case 1: Underproduction

Suppose $\tilde{q}$ is traded at price p^* :

- $CS = A$
- $PS = C$

Case 2: Equilibrium

What happens at q^* and p^* ?

- $CS = A + B$
- $PS = C + D$

Case 3: Overproduction

Suppose $\hat{q}$ is traded at price p^* :

- $CS = A + B - E$
- $PS = C + D - F$

Note: Trading beyond q^ creates negative surplus.*

The Equilibrium Principle

A market in equilibrium leaves no unexploited opportunities for individuals, but may not exploit all gains achievable through collective action.

Linear demand and supply curves

Equilibrium with linear supply and demand curves

Comparative Statics Analysis (Ceteris Paribus Analysis)

Aggregate Demand and Supply

Suppose that individual demand curves are given by $q_i^d(p)$, for $i = 1, \dots, I$ consumers; and individual supply curves are given by $q_k^s(p)$, for $k = 1, \dots, K$ producers/firms.

The **aggregate or market demand** is obtained by **horizontal summation** of all the individual demand curves — at price p , we add up all demanded quantities:

$$Q^D(p) = \sum_{i=1}^I q_i^d(p).$$

Similarly, the **aggregate or market supply** is obtained by **horizontal summation** of all the individual supply curves — at price p , we add up all supplied quantities:

$$Q^S(p) = \sum_{k=1}^K q_k^s(p).$$

Warning!

When doing the horizontal summation of the individual demand and supply curves to obtain the aggregate curves, you must always be careful to add them up at prices that lead to **positive quantities!**

Remember, individual demand and supply curves are only defined for positive prices and positive quantities!

